



Optimal inventory control for bottleneck isolation in general processes[☆]

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ABSTRACT

Optimal inventory control seeks to isolate the economic effect of bottlenecks and maximise the throughput of processes. This is challenging in complex topologies with disturbances causing shifting bottlenecks. Decentralised and model predictive control schemes have been proposed for bottleneck isolation of sequential processes. Although decentralised control schemes work well for sequential processes, they are difficult to apply to more complex topologies such as parallel arrangement of units, flow splits, mergers, and recycles that are common in the industry. In contrast, such multi-input multi-output systems can be naturally handled with model predictive control schemes. This work extends a preliminary model predictive control scheme in the literature to achieve bottleneck isolation in general process topologies. In particular, a seriatim amongst inventories and system outflows is created using weights in the objective function. Our approach is simple to implement and is shown to optimally isolate bottlenecks on a wide range of case studies and topologies.

1. Introduction

This paper studies how to design model predictive controllers for inventories in complex process systems. Inventories with various sizes are present in all process plants. Examples include levels and pressures in storage tanks, distillation columns, and reactors, etc. To maintain reliable operations, these inventories must be controlled within their minimum and maximum limits. If a bottleneck temporarily reduces the flow at some point, it is desirable to isolate the effect of the bottleneck, such that its effect on other parts of the process is minimal. In this case inventories act as buffer storages, such that the overall throughput can be maintained (or maximised), or to ensure continuous and smooth operations during periods of flow fluctuations. Several studies developed decentralised control structures for inventory control to maximise throughput while respecting inventory limits [1,2].

Generally there are two approaches to manage inventories [3]. One approach is to fix inventory set-points and use loose, averaging control to smooth flow transients. An alternative approach is to tightly control inventories at their limits (minimum or maximum), using them intentionally as buffers to maximise throughput [1,2,4]. When bottlenecks shift, this approach drains inventories after the bottleneck, before reducing steady-state throughput, thus effectively isolating the effect of the bottleneck. In our context, bottleneck refers to a constraint preventing further increase in steady-state flow [1]. The two approaches embody competing goals because bottleneck isolation cannot be achieved without inventory fluctuations. While the literature

traditionally has focused more on the former approach [5–7], the latter favours process economics in the form of throughput maximisation and this usually results in maximal profit.

The focus of most literature on decentralised inventory control has been on sequentially arranged inventories. However, industrial processes typically have complex process topologies with flow splits, mergers, and recycles. While it is certainly possible to handle complex inventory topologies with simple decentralised controllers, it can become tedious. Moreover, the resulting control structures and associated logic can be difficult to understand and diminish the benefit of simplicity that often comes with decentralised control structures.

Model predictive control (MPC) lends itself readily to challenging control problems with complex trade-offs. In particular, assigning weights to different control objectives makes it possible to design a controller prioritising the right inventories in the plant. However, aside from preliminary work [4], dedicated MPC schemes for inventory control are absent from the literature and there exist no systematic methods for selecting suitable MPC cost functions and weights for complex topologies.

In this paper we fill this gap and show how consistent inventory control can be achieved with MPC. We give a systematic procedure for weight selection in the MPC cost function. Our method can be used to design a model predictive controller that maximises throughput for general complex process networks consisting of flow mergers, splits, and recycles. The approach is applicable to any process with buffer

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capacities and the objective of maximising throughput. Examples include the pulp and paper industry [8], integrated process sites [9] such as refineries or chemical plants, water management systems [10], and wastewater purification systems [11,12].

Specifically, this work contains the following contributions:

1. We show how the inventory MPC of Turan et al. [4] can be extended to more complex process topologies (flow mergers, splits, and recycles),
2. We show how selected inventories or system outflows can be prioritised
3. We present a systematic and generalised algorithm that determines weights to control the MPC behaviour

The paper is structured as follows: Section 2 briefly reviews the essential background of the paper, including the inventory control problem, and MPC approach of Turan et al. [4], Section 3 proposes a generalised MPC approach for bottleneck isolation of complex processes, Section 4 presents several case studies in which the proposed approach is applied to various processes, and lastly, we end with a discussion and conclusion in Sections 5 and 6.

2. Background

2.1. Inventory control

Inventory control is an important part of process plant operation and has been extensively explored in the literature, e.g. [1,3,13]. Here, the term inventory refers to a unit with hold-up (pressure, liquid level, etc.). Controlling the hold-up in such a unit is equivalent to material balance control. Several studies have worked on guidelines for designing decentralised inventory control schemes [14–18]. These guidelines have been used in decentralised inventory control schemes for throughput maximisation [1,19].

Consider the following example: A process operates at maximum throughput with a bottleneck at the end of the process (system outflow or flow to consumer). Let us first assume the inventory set-points are fixed. Consider a temporary disturbance that introduces a new bottleneck further upstream. This fixed inventory approach immediately reduces the overall steady-state throughput of the system. If, instead the inventory set-points are treated as degrees of freedom, the influence of the bottleneck on process throughput can be delayed by selecting optimal inventory set-points based on operating conditions. In this example, inventory set-points downstream of the new bottleneck should be low, to drain inventories before ultimately reducing throughput. Once the temporary bottleneck is relieved, inventory set-points should be high to alleviate the effect of future disturbances. A key challenge with inventory control was formulated by Zotică et al. [1]¹:

ICC (Inventory Control Challenge:) *Use of intermediate storage for bottleneck isolation (containment): How to optimally select the inventory (level) set-points to maximise the time until a new bottleneck makes it necessary to decrease the throughput?*

Shinsky [19] and Zotică et al. [1] isolated bottlenecks for inventories in series using a decentralised scheme (bi-directional inventory control). When bottlenecks shift, the constraint control scheme maximises throughput insofar as inventories are within operational limits. Hence, the Inventory Control Challenge ICC is addressed through the use of inventory controllers with set-points at the maximum and minimum operational limits. Additionally, selectors were used to automatically shift the throughput manipulator together with shifting bottlenecks [1,2]. The throughput manipulator refers to the degree of

freedom used to set the steady-state flow in the system [20]. Consequently, inventories after the bottleneck are minimised while those before are maximised. Bernardino and Skogestad [2] later incorporated minimum flow constraints into this scheme. Alternatively, MPC schemes can be used for bottleneck isolation [4] and this will be discussed further in the next section.

2.2. MPC for inventory management

MPC has been widely used for the control of multi-input, multi-output processes with constraints. Inventories in complex topologies are examples of such processes. In these complex topologies, MPC is advantageous because it provides a framework for implicit rearrangement of input–output pairings under different operating conditions.

Turan et al. [4] presented MPC formulations for sequentially arranged inventories while maximising throughput and isolating bottlenecks. The solution of the MPC problem implicitly defines optimal inventory set-points and requires only a dynamic model of the inventories without any disturbance prediction.²

The MPC does not consider detailed process models, including nonlinearities and energy balances, and it is suitable as a higher-level optimiser in the control hierarchy [21]. It can manage production rates for various units, sub-units, and storage tanks, while utilising their buffer capacities for bottleneck isolation.

The formulations by Turan et al. [4] can be tuned to yield comparable performance to the bidirectional control scheme of Zotică et al. [1] and have been applied to control the levels of tanks in series [4]. However, they have not been extended to other more complex process topologies including splits, mergers, and recycles.

In this work, we adapt the MPC scheme to more general networked systems with multiple sources (inventories that receive flows into the system), sinks (inventories that have flows out of the system), and recycles. This is motivated by the need for prioritisation amongst multiple inventories and system outflows. Although this MPC scheme can be applied to any process with inventories, we will take the case of tanks connected in different topologies.

Let us consider a network of N_I inventories connected by N_F flows. We denote the set of real numbers by $\mathbb{R}$, the set of non-negative real numbers by $\mathbb{R}_+$, and the set of positive real numbers by $\mathbb{R}_{++}$. We assume these inventories to be tanks with levels, $h \in \mathbb{R}_+^{N_I}$, and constant cross-sectional areas, $a \in \mathbb{R}_{++}^{N_I}$. Inventory control in this case refers to level control for each tank. The discrete-time model for tank levels can be represented as follows using a volumetric basis:

$$ah(t_{k+1}) = ah(t_k) + M F(t_k) \quad (1)$$

$$M_{ij} = \begin{cases} 1 & \text{if } F_j \text{ enters vessel } i \\ -1 & \text{if } F_j \text{ exits vessel } i \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where $F \in \mathbb{R}_+^{N_F}$ is a vector of flows, and $M \in \mathbb{R}^{N_I \times N_F}$ is an incidence matrix describing the connectivity of the system.³

The control problem can be formulated as follows with the flow rate F as the manipulated variable and the tank height h as the controlled variable. Here, we assume that the flow rates are perfectly controlled using a flow controller. Thus, the MPC problem can be represented as follows:

$$\min J \quad (3)$$

$$h(t_{k+1}) = Ah(t_k) + BF(t_k) \quad (4)$$

$$h_{\min} \leq h(t_k) \leq h_{\max} \quad (5)$$

² In some cases, the disturbances may impact control performance. We refer to Turan et al. [4] for details.

³ Note that the model and the approach can easily be generalised to other inventories than tanks.

¹ They also formulated additional challenges that relate to decentralised control schemes.

$$0 \leq F(t_k) \leq F_{\max} \quad (6)$$

where J is an objective function (specified later), $h_{\min} \in \mathbb{R}_+^{N_I}$ and $h_{\max} \in \mathbb{R}_+^{N_I}$ are vectors that define the range of allowable levels, $F_{\max} \in \mathbb{R}_+^{N_F}$ defines the range of allowable flow rates, and N_k is the number of time steps. To present the tank dynamics in a general form as shown in (4), the matrices $A \in \mathbb{R}^{N_I \times N_I}$ and $B \in \mathbb{R}^{N_I \times N_F}$ are defined as the identity and the component-wise division of M/a respectively. Note that a non-zero minimum flow rate is not included for now as this could result in infeasibility. This assumption will be revisited later.

The bottleneck in the model is defined using $F_{\max}$, which we assume is a known, certain parameter. However, the shifts in bottleneck (i.e. changes to $F_{\max}$) are unknown a priori. In other words, a forecast of the bottleneck is not available to the MPC, although it could be included if available.

In previous work [4], the proposed objective of the MPC for a serial system of inventories is to maximise the system outflow and weighted inventories as follows:

$$J = - \sum_{k=0}^{N_k} \gamma^k \left(F_N(t_k) + \sum_{i=1}^{N_I} \alpha_i^T h_i(t_k) \right) \quad (7)$$

where $0 < \alpha_i^T < 1$ is the weighting factor for the inventories and F_N is the system outflow. The relative importance of the inventories is reflected in the magnitude of their weights which indicate the order in which inventories should be drained when a bottleneck shifts. The weights α_i are chosen such that throughput maximisation is prioritised over inventory build-up. The objective includes a discount factor, $0 < \gamma < 1$, which prioritises minimising the objective function at earlier time steps, and puts less importance on time steps further in the future. Without this discounting, all time steps are equally important resulting in “time symmetry” which leads to multiple solutions of the MPC problem, and thus a non-unique control law.⁴

3. Generalised inventory MPC for process networks

In this section, we discuss how the MPC cost (7) for serial inventories can be generalised to any process network by careful weighting of all the system outflows. We propose the following MPC formulation to handle more general process topologies:

$$\begin{aligned} \min J = & - \sum_{k=0}^{N_k} \gamma^k \left(\underbrace{\sum_{j \in \mathcal{O}} \alpha_j^O F_j(t_k)}_{\text{System Outflow Maximisation}} \right. \\ & \left. + \underbrace{\sum_{i=1}^{N_I} \alpha_i^I a_i h_i(t_k)}_{\text{Inventory Maximisation}} - \underbrace{R}_{\text{Sum of Penalty terms}} \right) \end{aligned} \quad (8)$$

$$h(t_{k+1}) = Ah(t_k) + BF(t_k) \quad (9)$$

$$h_{\min} \leq h(t_k) \leq h_{\max} \quad (10)$$

$$0 \leq F(t_k) \leq F_{\max} \quad (11)$$

where $1 < \alpha_j^O$ is the weighting factor for system outflows, F_j . R refers to the sum of various penalty terms such as constraint violations (e.g. inventory limits, minimum flow limits) or input regularisation. These will be discussed further below. The cost function (8) maximises throughput and inventory levels. The following guidelines govern its application:

1. Maximising outflows should take priority over maximising inventories, since the latter are meant to act as buffers to mitigate throughput reduction. This prioritisation should be reflected in the choice of the weighting parameters α_j^O and α_i^I , which are related to system outflow and inventory maximisation respectively. In other words, $\min_{j \in \mathcal{O}} \alpha_j^O > \max_{i \in \mathcal{I}} \alpha_i^I$. In this study, 1 is chosen as a threshold value⁵ such that $\alpha_j^O > 1 > \alpha_i^I$.
2. The magnitudes assigned to α_j^O must be large enough such that the term for system outflow maximisation is greater than the term for inventory maximisation in (8).
3. A preference exists among outflows (akin to multiple products of a process plant). To reflect this, α_j^O should be assigned a unique value.⁶
4. Inventory maximisation depends on the inventory weights α_i^I . They should be assigned based on the relative importance of an inventory within the network.
 - (a) An inventory located closer to a consumer should have higher priority for maximisation. When bottlenecks shift, this ensures buffer storages are drained (starting from the inventory furthest from the consumer) before system outflows are cut.
 - (b) An inventory connected to a more important consumer should have a higher priority for maximisation.

The key challenge with generalising the MPC formulation to different and more complex process topologies is the appropriate selection of inventory weights considering the aforementioned concepts. Below we show how this can be achieved in a systematic fashion.

3.1. Calculating inventory weights

Consider a network consisting of process inventories, consumers (nodes that take in process outflows), and producers (nodes that supply process inflows). These nodes are assigned a label of either storage S , producer P , or consumer C . Thus, the total number of nodes is given by $N_N = |S| + |P| + |C|$, where there are N_F connections (flows) between the nodes. In total, there are N_N nodes, of which N_I nodes are of label S . In total, there are N_E connections between all the nodes where $N_F = N_E$. Connectivity between the nodes is described by an incidence matrix M^* .⁷ Flows from a storage node to a sink node are labelled system outflows. Let $\mathcal{O}$ be the sets of the indices of these nodes.

With the process network set-up as described above, each storage node's weight can be determined based on its proximity to consumers and the importance of the consumers it is connected to. We propose Algorithm 1 to calculate inventory weights.

We will use an example process network, as shown in Fig. 1, to illustrate the calculation of inventory weights. The information required to apply the algorithm is as follows:

- The network consists of 5 nodes (N_N) in total, 3 storage nodes, 1 producer node, and 2 consumer nodes. There are 5 edges (N_E) that represent 5 flows (N_F) between the nodes. Connectivity can be described using the incidence matrix,

$$M^* = \begin{bmatrix} P_1 \\ S_1 \\ S_2 \\ S_3 \\ C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

⁵ Other threshold values could be chosen.

⁶ Unique system outflow weights avoids non-unique solutions.

⁷ Note that the incidence matrix M^* is different from the incidence matrix M because it includes the sink and source nodes.

⁴ Here, the cost function and the dynamic tank model are linear, rendering this a convex optimisation problem. However, the method can also be applied to nonlinear models, too.

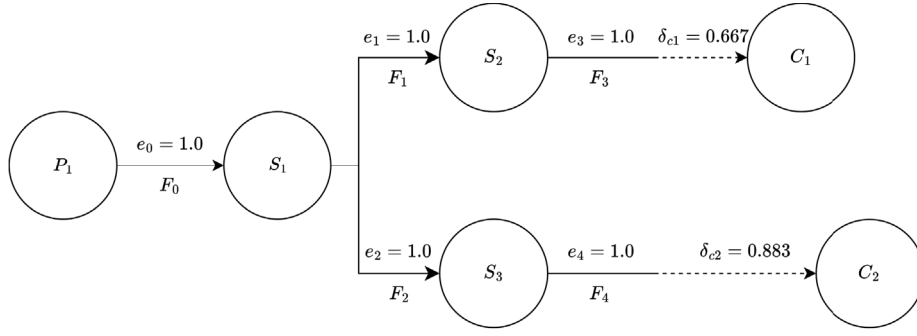


Fig. 1. Process network example.

Algorithm 1 Basic algorithm to systematically find inventory weights given outflow weights

Require: Network of nodes, index list of nodes labelled storage $s \in S$ and consumer $c \in C$, Vector of flows F , index list of flows labelled outflows $j \in \mathcal{O}$ and incidence matrix M^*

Require: edge weights, $e \in \mathbb{R}_{++}^{N_F}$. Default: $e = 1$

Require: For each $j \in \mathcal{O}$ assign a weight $1 < \alpha_j^O \in \mathbb{Z}$. A larger value implies a preference for this stream over another.

0: Allocate: $\delta_c \in \mathbb{R}_{++}^{|C|}$, $d_{c,s} \in \mathbb{R}_{++}^{|C| \times |S|}$, $\bar{d}_s \in \mathbb{R}_{++}^{|S|}$, $\alpha_s^I \in \mathbb{R}_{++}^{|S|}$
 1: $\delta_c \leftarrow \text{consumer_priority}(M, \alpha, c)$ for each consumer c
 2: $d_{c,s} \leftarrow \text{shortest_path}(s, c)$ for each consumer c and storage s
 3: $\bar{d}_s \leftarrow \min_{c \in C} (d_{c,s} + \delta_c)$ for each storage s
 4: $\alpha_s^I \leftarrow 1/\bar{d}_s$ for each storage s
 5: **return** α^I

- Each edge is assigned an edge weight, e , representing the distance between two nodes. A larger value implies a preference for other streams over this stream. This is because a larger edge weight situates a storage node further away from consumers, thus decreasing its relative importance in the system. For pedagogical reasons, we assume that all edges in the network have a weight of 1. This assumption will be relaxed in Section 3.1.1.
- System outflows are assigned weights, α_j^O . These should have a magnitude greater than 1 based on preferred production pathways. A larger value implies a preference for the stream over another because it is then prioritised for maximisation. This process has 2 system outflows, F_3 and F_4 . Let us assume that the former is of greater importance. Therefore, the system outflow weight for F_3 is assigned a higher value ($\alpha_3^O = 1.5$) than F_4 ($\alpha_4^O = 1.2$).

With this information, we can apply Algorithm 1 to calculate inventory weights as follows:

Step 1 assigns larger inventory weights to storage nodes connected to higher priority consumers. Consumers with greater priority (i.e. higher system outflow weight) are situated closer to the storage nodes they are connected to. This is done by adding an incremental distance, δ_c , to the edge weight connecting storage and consumer nodes. Consumer nodes with greater priority have smaller values for δ_c . This incremental distance, δ_c , is equal to the inverse of the system outflow weight. For the example in Fig. 1, $\delta_c = \begin{bmatrix} 0.667 \\ 0.883 \end{bmatrix}$.

Step 2 determines the shortest distance, $d_{c,s}$, between each storage node and consumer node. This is equal to the summation of the edge weights. For the example in Fig. 1, $d_{c,s} = \begin{bmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \end{bmatrix} = \begin{bmatrix} 2 & 1 & \infty \\ 2 & \infty & 1 \end{bmatrix}$.

Step 3 calculates $\bar{d}_s$ which represents the distance from each storage node to the closest consumer node it is connected to. First, the consumer priorities are accounted for by adding δ_c to $d_{c,s}$. Next, the shortest distance from each storage node to any one of the consumer nodes is selected. $\bar{d}_s$ reflects both the relative position of a storage node in the network and the importance of the consumer nodes connected to it. For the case in Fig. 1, $\bar{d}_s = \min_c (d_{c,s} + \delta_c) = \min \begin{bmatrix} 2.667 & 1.667 & \infty \\ 2.883 & \infty & 1.883 \end{bmatrix} = [2.667 \ 1.667 \ 1.883]$.

Step 4 inverts the distance metric in Step 3 to obtain inventory weights that are smaller than the system outflow weights (α_j^O). Thus, the inventory weights in Fig. 1 are $\alpha_s^I = \begin{bmatrix} \frac{1}{2.667} & \frac{1}{1.667} & \frac{1}{1.883} \end{bmatrix} = [0.375 \ 0.599 \ 0.531]$. This implies that storage node 2 is the most important followed by storage node 3 and 1.

In summary, Algorithm 1 generates weights α^I that can be used in the generalised MPC objective in Eq. (8). Below we discuss further extensions of Algorithm 1 for systems with preferences in internal paths (parallel branches that are split or merged), systems with recycle, and systems where a minimum flow rate is desired.

3.1.1. Edge weights

Algorithm 1 requires the user to pre-assign edge weights. In Section 3.1, we assumed transfer between units to be equally preferred (the edge weights are all 1). If some transfers are preferred less over others they should be assigned a larger edge weight. Essentially, larger edge weights locate storage units further away from the sinks, thus lowering the importance of the storage unit. Below we provide guidelines on how one can assign edge weights.

1. Let us take the case of storage nodes arranged in parallel, that are equidistant from the same consumer. Edge weights for outlet streams from these storage nodes should be unique to avoid multiple solutions (This is illustrated in Examples 2 and 5 below).
2. If a storage node exists in recycle streams, its outlet stream should have an edge weight reflecting the node's relative importance in the process network (see Example 3).
3. If a storage node is known to have a different priority based on process knowledge, its outlet stream should have an edge weight reflecting the node's relative importance in the process. In other words, edge weights can be used to introduce process information.

In the following sections, we will discuss additional elements of the MPC formulation for certain special cases of process topologies.

3.2. Input regularisation

Certain process structures present multiple solutions even when all outflow weights and inventory weights are unique. For example, there

may be multiple steady-state solutions when parallel streams are not at their maximum flow rate (see [Examples 2 and 5](#)). Input regularisation can be applied to prevent the optimiser from repeatedly changing strategies or making excessive movements in the MVs. It can be implemented as a linear soft constraint using slack variables as shown in [\(12\)](#) and [\(14\)](#). The penalty for MV movement is a tuning parameter that must be chosen carefully depending on process requirements.

$$F_j(t_k) - F_j(t_{k-1}) \leq s_{1,j}(t_k), \forall t_k \in 1 : N, \forall j \in 0 : N_F \quad (12)$$

$$-(F_j(t_k) - F_j(t_{k-1})) \leq s_{2,j}(t_k), \forall t_k \in 1 : N, \forall j \in 0 : N_F \quad (13)$$

$$R_{input_reg} = \sum_{k=1}^N \sum_{j=0}^{N_F} \rho_{1,j}(s_{1,j}(t_k) + s_{2,j}(t_k)) \quad (14)$$

3.3. Processes with recycle

Recycle streams are common in the process industry. From the perspective of throughput maximisation, it is desirable to minimise the amount recycled as it flows in the opposite direction as the outflow. However, recycles are necessary for various reasons, and the recycle ratio may not even be a degree of freedom. For example the magnitude of a recycle stream could be governed by physics as in a distillation column or a flash tank. The recycle could also be used to manage operational concerns. For example, compressors have minimum flow rate constraints to avoid unstable operation. A recycle loop around the compressor is used to manage this constraint. It may also be that a recycle flow may be constrained to a fixed value, as in Luyben [\[16\]](#), Larsson et al. [\[22\]](#), or set to a fixed ratio ([Example 3](#), to model situations where the recycle is governed by physics) or used to ensure satisfaction of minimum flow constraints ([Example 6](#)). As recycles can be due to a myriad of factors, we here use constraints to model the recycle in a simplified manner.

3.4. Specification of a minimum flow rate

Typically, industrial processes have minimum turndown constraints which prevent control systems from automatically cutting off certain flows. Hence, operational constraints might require a minimally allowable flow rate, F_{min} , for certain streams. Bernardino and Skogestad [\[2\]](#) have previously considered this problem for the case of serially connected vessels. One cannot simply include F_{min} as a constraint as this can lead to an infeasible problem. Specifically it can be impossible to satisfy both the constraint on the minimum flow rate and the constraint on the minimum level.

By including both (minimum flow and level limits) as soft constraints, the trade-off between these two constraints can be made explicit. We consider the use of the following penalty term in the objective:

$$F_{j,min} \leq s_{3,j}(t_k) + F_j(t_k), \forall t_k \in 0 : N, \forall j \in 0 : N_F \quad (15)$$

$$R_{min_flow} = \sum_{k=0}^N \sum_{j=0}^{N_F} \rho_{2,j} s_{3,j}(t_k) \quad (16)$$

where $\rho \in \mathbb{R}^{N_F}$ is a penalty parameter. As the objective and penalty term are linear, one can explicitly specify the prioritisation of this constraint and the objective. Generally, one will prefer satisfaction of the constraint over the objective and thus one should specify ρ to satisfy the constraint:

$$\max(\alpha_j^0) < \rho \quad (17)$$

If one chooses to include other soft constraints such as constraints on the inventory rate of change, there could be some preference in which constraints should be satisfied. In such a case, the most prioritised constraint should have the largest ρ , and so on.

4. Case studies

We will now apply the proposed approach to three common process structures - (1) Units in Parallel, (2) Units with Recycle, and (3) Units in a Process Network. A total of six examples are presented (see [Appendix A](#) for [Examples 5 and 6](#)), each with slight variations in process structure. The parameters are assumed as follows, unless stated otherwise within an example. Outflow weights and edge weights are 1.5 and 1.0 respectively. Minimum and maximum tank level limits are 10% and 90% of the total tank height, respectively. At the initial condition, tank levels are at their maximum limits and flow rates are at steady-state, F_{ss} . Based on the process structure (incidence matrix), outflow weights and edge weights, [Algorithm 1](#) is applied to calculate the inventory weights. All tanks sizes, maximum flow rates, steady-state flow rates, and inventory weights are annotated on figures showing the process structures for each example ([Figs. 2, 4, 7, 9, A.14, and A.11](#)). Note that the annotations for maximum flow rates are enclosed in boxes with dashes lines while those for steady-state flow rates are enclosed in boxes with solid lines. A summary of all the user-defined parameters can be found in [Appendix B](#). Each example shows the MPC response to a shift in the process bottleneck and/ or a tank leak.

4.1. Units in parallel

Example 1 (Tank arrangement with flow split). Consider a configuration of four tanks with a flow split as shown in [Fig. 2](#) with process bottlenecks at F_3 and F_5 . Here we consider that processing through tank 3 is preferred over tank 4. Hence, F_3 ($\alpha_3^0 = 1.5$) is assigned a higher outflow weight than F_5 ($\alpha_5^0 = 1.2$) and tank 3 has a larger inventory weight than tank 4. This implies that maximising the inventory of tank 3 takes priority over tank 4. Although tanks 3 and 4 are in parallel, they do not require unique edge weights for their outlet streams. This is because the stream outflow weights create a distinction in the inventory weights of those tanks.

We simulate the system with a reduction in $F_{0,max}$ between 10 min and 60 min such that F_0 is now the bottleneck ($F_{0,max} = 0.5 \text{ m}^3/\text{min}$). Additionally, an uncontrolled leak of $0.05 \text{ m}^3/\text{min}$ from tank 3 is introduced at 30 mins. [Fig. 3](#) shows the MPC response when the bottleneck is shifted to F_0 . The tanks further away from the outflows are drained first. Between tank 3 and 4, the latter is drained first. When all tanks are at their lower limit, F_5 is reduced instead of F_3 due to its lower outflow weight. Note that F_3 is as close to its maximum as possible. It is lower than $F_{3,max}$ by $0.05 \text{ m}^3/\text{min}$ because it is the only input available to manage the leak from tank 3.

Example 2 (Tank arrangement with flow merger). Now consider a configuration of four tanks with a flow merger as shown in [Fig. 4](#) with the process bottleneck at F_5 . Here, we consider that maximising the inventory of tank 2 is preferred over tank 1. Hence, an edge weight of 1.2 is assigned to F_1 . Thus, tank 2 has a higher inventory weight than tank 1.

We simulate the system with a reduction in $F_{0,max}$ and $F_{2,max}$ between 10 min and 60 min such that F_0 and F_2 are now the new bottlenecks ($F_{0,max} = F_{2,max} = 0.25 \text{ m}^3/\text{min}$). Additionally, an uncontrolled leak of $0.05 \text{ m}^3/\text{min}$ from tank 1 is introduced at 30 min. [Fig. 5](#) shows the response of the MPC when the bottleneck is shifted. Since both $F_{0,max}$ and $F_{2,max}$ have been lowered, tanks 1 and 2 must drain. However, since tank 1 is assigned a lower inventory weight, it is drained faster than tank 2. Additionally, F_1 is lowered after 30 min to manage the leak in tank 1. When the nominal maximum flow rates are restored, the inventory of tank 2 rises faster than that of tank 1 due to its higher weight. It is important to note that this problem presents multiple solutions since the steady-state flows through tank 1 and tank 2 are not at their maximum values. Hence, linear input regularisation is used with a small penalty of 10^{-3} to prevent excessive MV movement. The dynamics in

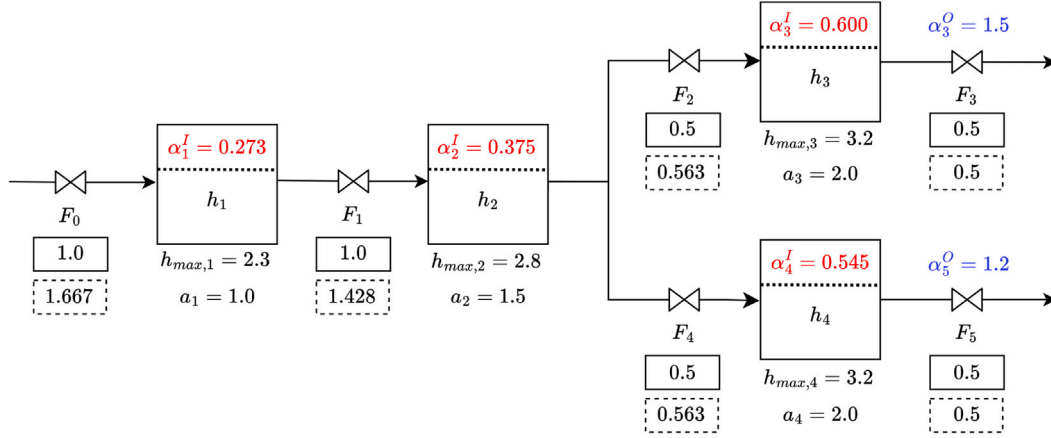


Fig. 2. Example 1, tank arrangement with flow split. Maximum flow rates in box with dashed lines, steady-state flow rates in box with solid lines.

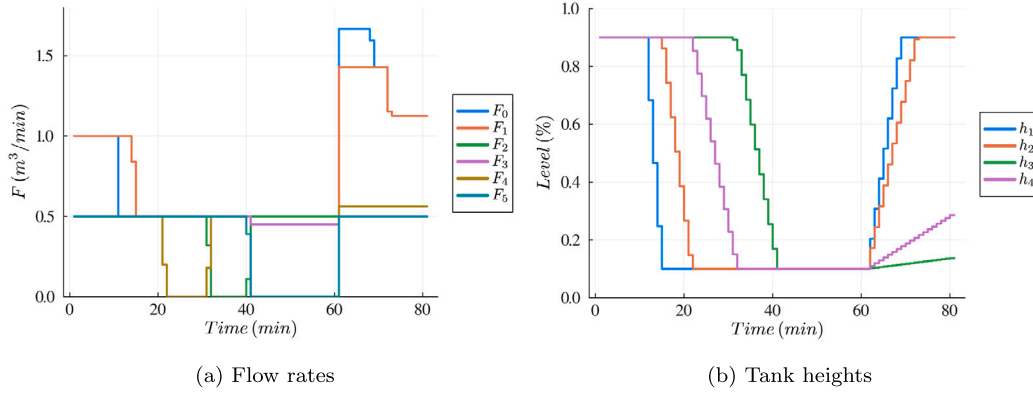


Fig. 3. MPC Response, Example 1 - tank arrangement with flow split.

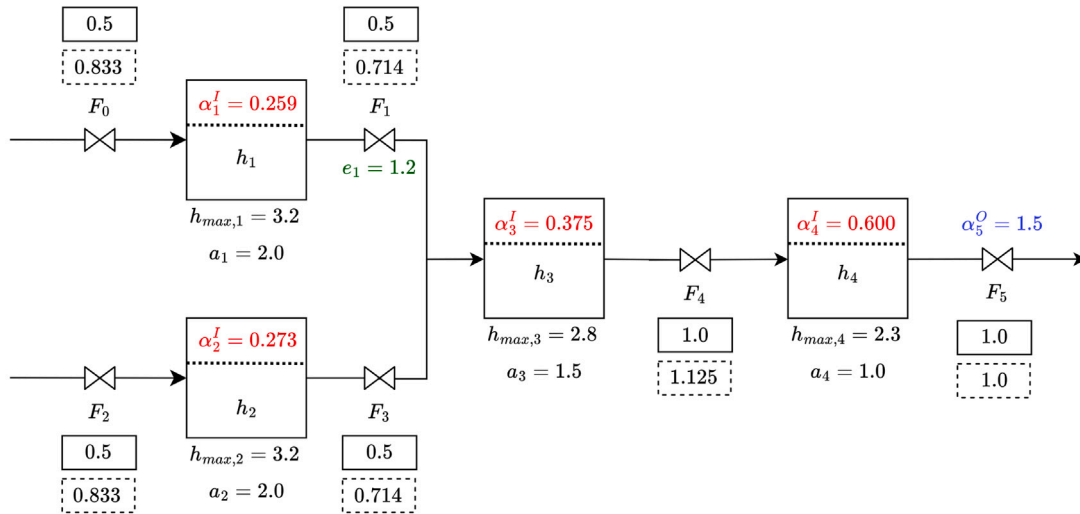


Fig. 4. Example 2, tank arrangement with flow merger. Maximum flow rates in box with dashed lines, steady-state flow rates in box with solid lines.

the MPC solution depends on the magnitude of this penalty. One needs to carefully consider process objectives and operational constraints to set this value. Fig. 6 shows how the solution changes (i.e. tanks 1 and 2 are drained together) when a larger input regularisation penalty of 10^{-2} is applied.

4.2. Units with recycle

Example 3 (Water purification process). This example is inspired by the reverse osmosis process used for wastewater purification presented in Bortolini et al. [11]. An overview of the process structure used in

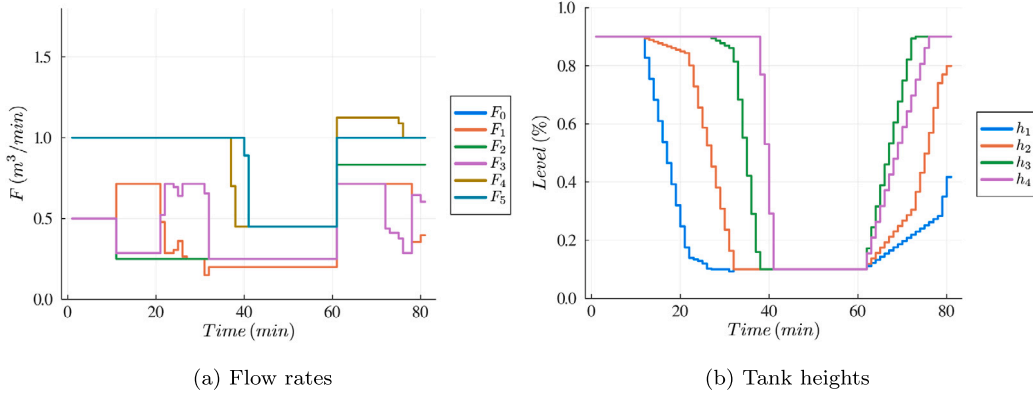


Fig. 5. MPC Response, Example 2 - tank arrangement with flow merger and applied input regularisation penalty of 10^{-3} .

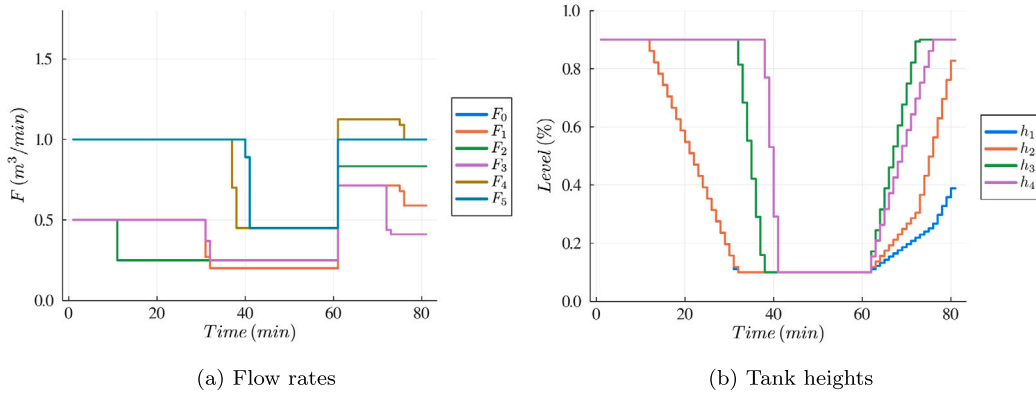


Fig. 6. MPC Response, Example 2 - tank arrangement with flow merger and applied input regularisation penalty of 10^{-2} .

this example is shown in Fig. 7. The process begins with the collection of wastewater from the cooling tower or other units into an input tank. Wastewater is then fed into a pre-filtration unit, with a processing efficiency of approximately 90% [11]. A buffer tank collects the filtered water before feeding it to the reverse osmosis (RO) and UV filtration unit. The RO process operates at an efficiency of 60% to produce pure water [11]. Retentate from the RO process is then recycled back to the input tank. In this example, slight modifications have been made to expand the inventory network. We have included a buffer tank within the recycle stream based on another process flow diagram for reverse osmosis [23]. We also added buffer tanks for individual wastewater streams (tanks 1 and 2).

The pre-filtration and RO processes are assumed to operate at steady state. Hard constraints for flow ratios are introduced to model process efficiency. The ratio of F_5 is fixed to be equal to 90% of F_4 . Similarly, the ratio of F_7 and F_8 is fixed to be equal to 60% and 40% of F_6 respectively. Here, the buffer tank in the recycle stream is considered to be of least importance relative to all other inventories. Hence, the outflow from the recycle tank, F_9 , is assigned an edge weight of 1.8. This ensures that tanks 5 and 1 do not have the same inventory weight. Furthermore, since tanks 1 and 2 are in parallel, the latter is assumed to have greater priority. Thus, an edge weight of 1.2 is applied to F_2 to reflect that.

Initially, the process bottleneck is at F_7 . We simulate the system with a reduction in $F_{1,max}$ between 10h and 60h such that it is now the bottleneck ($F_1 = 20.0 \text{ m}^3 \text{ h}^{-1}$). As shown in Fig. 8(b), the first tank to drain is that in the recycle stream because it has the smallest inventory weight. This is done by reducing F_3 to $20.0 \text{ m}^3 \text{ h}^{-1}$, maintaining throughput in the processes, and increasing the recycle stream, F_9 by

$1.9 \text{ m}^3 \text{ h}^{-1}$. Subsequently, the remaining buffer tanks are drained in sequential order based on their proximity to F_7 (tank 2, tank 3, and tank 4). When all the tanks are drained to their lower limit, pure water flow (F_7) is lowered. When the bottleneck on F_1 has been relieved, the tanks are once again filled to their maximum limits.

4.3. Units in a process network

Example 4 (Process network). In this example, we incorporate all the elements discussed thus far: flow splits, mergers, and recycles. Fig. 9 provides an overview of the process structure, including the maximum and steady-state flows for each stream.

This process is a supplier to two consumers through streams F_9 and F_{10} . Based on the maximum flow rates for each stream, the process bottlenecks are at F_9 and F_{10} . Here we consider that F_9 has higher priority for maximisation than F_{10} . Hence, F_9 ($\alpha_9^0 = 1.5$) is assigned a higher outflow weight than that of F_{10} ($\alpha_{10}^0 = 1.2$). The process also consists of two sets of inventories arranged in parallel. The first set consists of tanks 1 and 2. We consider tank 2 to have greater importance than tank 1. Hence, F_2 is assigned an edge weight of 1.2. The second set consists of tanks 4, 5, and 6. Since tank 6 is connected directly to a consumer, its outflow weight increases its importance for maximisation. Between tanks 4 and 5, the former is assumed to have greater priority. Thus, F_7 is assigned an edge weight of 1.1 and F_8 that of 1.2. The process also consists of a recycle stream intended to manage a minimum flow constraint of $0.35 \text{ m}^3/\text{min}$ on F_2 . The outflow from tank 8 in the recycle (F_{12}) is assigned an edge weight of 1.2 to ensure tank 8 has the least priority for maximisation. Note that tank 3

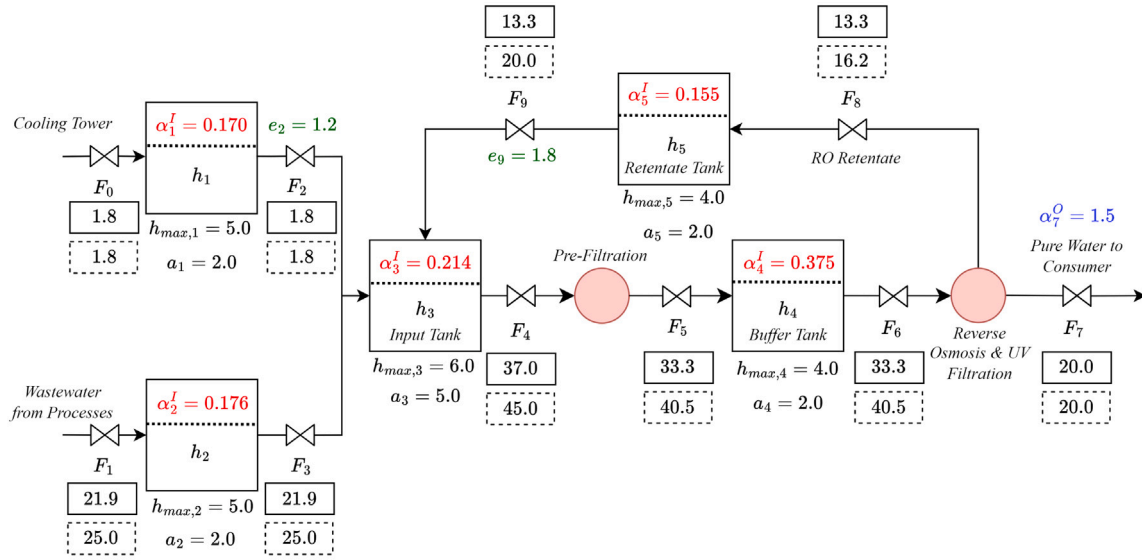


Fig. 7. Example 3, Wastewater purification process. Maximum flow rates in box with dashed lines, steady-state flow rates in box with solid lines.

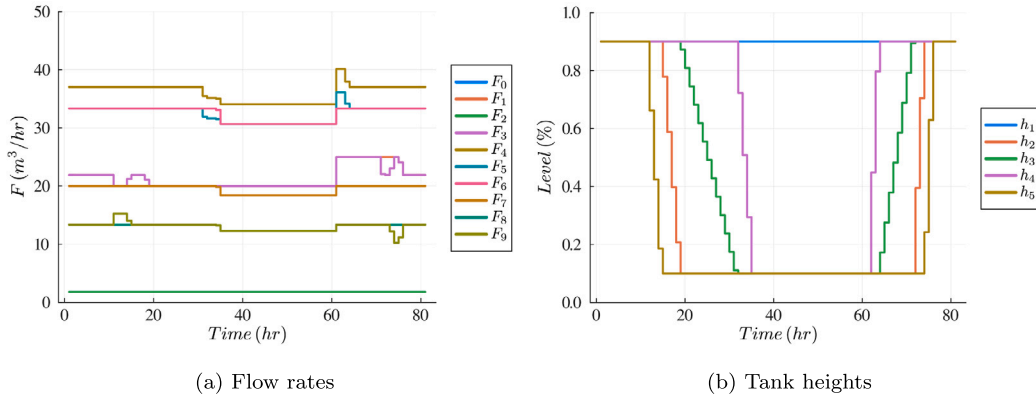


Fig. 8. MPC Response, Example 3 - Wastewater purification process.

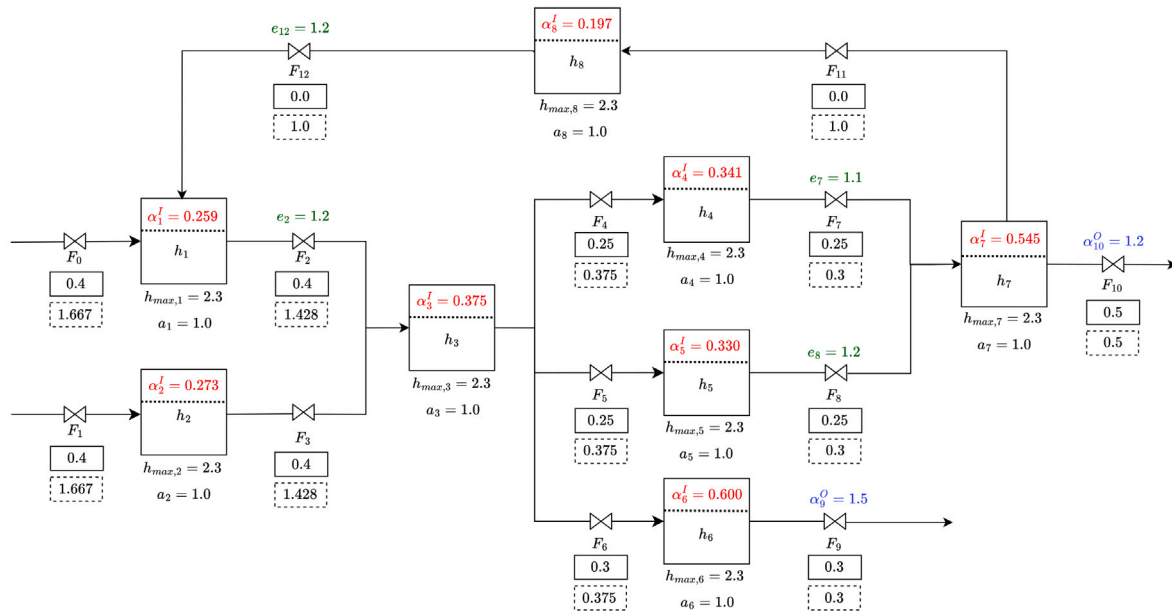


Fig. 9. Example 4, tank arrangement with flow splits, mergers, and recycle. Maximum flow rates in box with dashed lines, steady-state flow rates in box with solid lines.

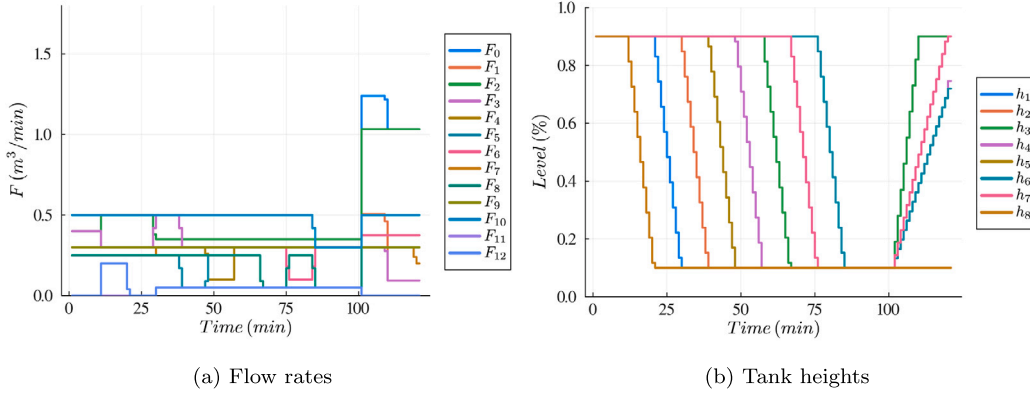


Fig. 10. MPC Response, Example 4 - tank arrangement with flow splits, mergers, and recycle.

has a higher priority for maximisation than tanks 4 and 5 because it is connected to a more important consumer (F_9).

We consider a reduction in $F_{0,max}$ and $F_{1,max}$ between 10 min and 100 min such that F_0 and F_1 are the new bottlenecks ($F_0 = F_1 = 0.3 \text{ m}^3/\text{min}$). Input regularisation with a penalty of 10^{-3} is implemented to prevent excessive MV movement. Since the steady-state flow rates in some streams are much lower than their maximums, a soft constraint is imposed on the inventory rate of change to prevent sudden changes in inventory. The inventory rate of change is constrained to $0.1h_{i,max}$. The MPC response is shown in Fig. 10. Based on Fig. 10(b), it is clear that the inventories are drained in the correct order as dictated by the inventory weights. A brief explanation of how each tank is drained is given below.

1. Tank 8 is drained by increasing F_{12} from $0 \text{ m}^3/\text{min}$ to $0.2 \text{ m}^3/\text{min}$.
2. Tank 1 is drained by keeping F_2 higher (at $0.5 \text{ m}^3/\text{min}$) and F_3 lower (at $0.3 \text{ m}^3/\text{min}$). Note that F_{12} is closed after tank 8 has drained.
3. Tank 2 is drained by keeping F_2 as low as possible (at $0.35 \text{ m}^3/\text{min}$) and F_3 high (at $0.5 \text{ m}^3/\text{min}$). Note that the recycle flows F_{11} and F_{12} are opened (by $0.05 \text{ m}^3/\text{min}$) to ensure F_2 is at its minimum flow constraint. Additionally, F_4 and F_7 are increased by $0.05 \text{ m}^3/\text{min}$ to manage the additional flow brought in through the recycle streams.
4. Tank 5 is drained by reducing F_5 by $0.2 \text{ m}^3/\text{min}$. Note that F_3 is reduced such that it is equal to F_1 (at $0.3 \text{ m}^3/\text{min}$) after tank 2 has completed draining.
5. Tank 4 is drained by reducing F_4 by $0.2 \text{ m}^3/\text{min}$. Note that F_5 is increased back up to $0.25 \text{ m}^3/\text{min}$ once Tank 5 has drained.
6. Tank 3 is drained simply by increasing F_4 back up by $0.2 \text{ m}^3/\text{min}$ once tank 4 has completed draining.
7. Tank 7 is drained by reducing F_8 and F_5 by $0.2 \text{ m}^3/\text{min}$.
8. Tank 6 is drained by reducing F_6 by $0.2 \text{ m}^3/\text{min}$. Note that F_8 and F_5 are increased back up by $0.2 \text{ m}^3/\text{min}$ when tank 7 has completed draining.
9. Since F_9 is prioritised for maximisation and all tanks have been drained, throughput in F_{10} is reduced by $0.2 \text{ m}^3/\text{min}$.

5. Discussion

5.1. Solution multiplicity

One of the key challenges with implementing the MPC for a large network is solution multiplicity. Large process networks present with several MVs that can be used to manage bottleneck shifts. It is undesirable for the optimiser to switch frequently between different strategies.

This could result in excessive and large MV movements. One approach to manage this is input regularisation as shown in Examples 2 and 5. The penalty weight used for input regularisation must be carefully tuned. Too large a penalty results in sluggish responses while too small a penalty leads to excessive MV movement. Another strategy is to incorporate constraints which can narrow down the feasible solution space. For example, equality constraints were used in Example 5 to balance the outflows from the 3 parallel tanks (see Fig. A.13). Equality constraints are typically applicable to processes with parallel units that require an equal balance of the load. Essentially, these parallel inventories and flows have equal importance. Since these equality constraints could interfere with obtaining the optimal solution, their criticality and allowable constraint violation must be carefully considered.

5.2. Penalty weights

The examples in this study include several constraints. These include minimum and maximum inventory limits, minimum flow limits, input regularisation, and equality constraints. This list is not exhaustive. If the problem modelled with hard constraints is feasible, then by selecting appropriately large penalty weights, the exact solution is recovered [24]. If the problem modelled with hard constraints is infeasible, then the penalty weights define a hierarchy amongst the soft constraints, akin to a multi-objective optimisation problem. See Sokaert and Rawlings [25]. In general, a larger penalty weight is used for a constraint of greater importance. Since the hierarchy of constraints is process-specific, a clear understanding of the process and objectives is required.

5.3. Profitability with inventory management

Our MPC formulation does not explicitly incorporate profit in the objective function. However, it is implicitly incorporated due to the assumption that maximising throughput is economically preferred. The tuning parameters such as edge, penalty, and system outflow weights present an opportunity for the user to implicitly incorporate process economics into the objective function. It is possible to also weight other flows within the system if they are deemed to have a high effect on profit.

5.4. Conflicting goals in inventory control

The MPC formulation presented in this study targets throughput maximisation, thus maximising inventories before the bottleneck and minimising those after the bottleneck. To truly maximise throughput (and thus profit), the inventory limits in the MPC must be as close

as possible to the true operational limits, requiring tight inventory control. The objective of throughput maximisation conflicts with the desire for flow smoothing during transient scenarios (i.e. loose level control), refer to Skogestad [3]. However, if one sought to incorporate averaging inventory control within the throughput maximisation MPC presented here, it could be done in two ways depending on how the lower regulatory control layer is configured.

If the MPC were to output optimal inventory set-points to PI level controllers, a slower PI controller tuning and more conservative inventory limits in the MPC could be used. On the other hand, if the MPC were to output flow set-points, the same could be achieved by increasing input regularisation for flow set-points and selecting more conservative inventory limits in the MPC. Additionally, one could apply adaptive tuning for PI level controllers or input regularisation (by varying penalty weights), depending on the magnitude of constraint violations on the inventory limits. Large constraint violations beyond a threshold would apply more aggressive controller tunings.

6. Conclusion

This study has extended the MPC formulation for inventory management presented by Turan et al. [4] to various process topologies including flow splits, mergers, and recycles. We have shown how priorities can be assigned to maximise system outflows and inventories using appropriate weights. Given the challenges associated with assigning inventory weights for large networks, an algorithm was developed to automatically calculate them. These weights maximise system outflows and inventories based on a desired seriatim even as disturbances cause bottlenecks to shift. The proposed approach is shown to isolate bottlenecks and maximise throughput in several case studies.

One of the key challenges with applying this MPC to a large process network is solution multiplicity. This results mainly from parallel arrangement of inventories. We have shown how this could be avoided through the application of input regularisation or a fixed ratio for flows. Future work can consider different forms of inventories such as pipeline pressures or component balances. Additionally, one could incorporate priorities amongst producers for cases where it is not economically desirable to reduce feed from certain producers.

CRediT authorship contribution statement

Archana Kumaraswamy: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Evren Mert Turan:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Johannes Jäschke:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table B.1

Example 1, parameters.

Tank Areas, m ²	[1.0 1.5 2.0 2.0]
Tank Heights, m ²	[2.3 2.8 3.2 3.2]
Maximum Flow Rates, F_{max} , m ³ /min	[1.667 1.428 0.563 0.500 0.563 0.500]
Steady State Flow Rates, F_{ss} , m ³ /min	[1.0 1.0 0.5 0.5 0.5 0.5]
Edge Weights, e	–
Outflow Weights, α^O	$\alpha_1^O = 1.5$, $\alpha_5^O = 1.2$
Inventory Weights, α^I	[0.273 0.375 0.600 0.545]

Table B.2

Example 2, parameters.

Tank Areas, m ²	[2.0 2.0 1.5 1.0]
Tank Heights, m ²	[3.2 3.2 2.8 2.3]
Maximum Flow Rates, F_{max} , m ³ /min	[0.833 0.714 0.833 0.714 1.125 1.000]
Steady State Flow Rates, F_{ss} , m ³ /min	[0.5 0.5 0.5 0.5 1.0 1.0]
Edge Weights, e	$e_1 = 1.2$
Outflow Weights, α^O	$\alpha_5^O = 1.5$
Inventory Weights, α^I	[0.259 0.273 0.375 0.600]

Appendix A. Additional examples

We include two additional examples here demonstrating the application of input regularisation (see Example 5) and minimum flow constraints (see Example 6) to manage potential operational requirements.

Example 5 (Tank arrangement with flow split and merger). We will now consider a configuration of five tanks including both a flow merger and split as shown in Fig. A.11 with the process bottleneck at F_7 . Processing is preferred through tank 2, followed by tank 3 and then tank 4. Thus, edge weights of 1.1, 1.2, and 1.3 are applied to F_4 , F_5 , and F_6 . This implies that maximising the inventory of tank 2 is most important followed by tank 3 and then tank 4. We simulate the system with a reduction in $F_{0,max}$ between 10 min and 60 min such that F_0 is now the new bottleneck ($F_{0,max} = 0.5$ m³/min). Fig. A.12 shows the response of the MPC when the bottleneck is shifted. As expected, tank 1 is drained first, followed by tanks 4, 3, 2, and 5. Input regularisation is also used in this example to prevent excessive MV movement. A simpler solution as opposed to the MPC solution in Fig. A.12(a) is to drain tanks 2 to 4 only using their inlet flows. Such a solution can be obtained through the enforcement of soft equality constraints for F_4 , F_5 and F_6 (see Fig. A.13(a)).

Example 6 (Tank arrangement with recycle stream). We now consider three tanks in series with a recycle stream from tank 3 to tank 2. The flow through F_2 could be viewed as analogous to a compressor system, where a recycle stream is used to ensure minimum flow through the compressor. A soft constraint is enforced on the minimum flow of F_2 ($F_{2,min} = 0.75$ m³/min). The recycle stream is used to manage the minimum flow constraints. Initially, the process bottleneck is at F_4 and there is no flow through the recycle.

We again consider a reduction in $F_{0,max}$ between 10 min and 60 min such that F_0 is now the bottleneck ($F_{0,max} = 0.5$ m³/min). Similar to Example 1, the tanks are drained beginning from the tank furthest away from the consumer (see Fig. A.15). Additionally, the recycle stream is opened to avoid violation of the minimum flow constraint for F_2 . Once the bottleneck has been relieved and F_2 returns to its nominal value, the recycle stream closes.

Appendix B. Case study parameters

In Tables B.1–B.6, values for all the parameters used in the examples have been summarised. Edge weights are not specified unless they are different from the default value of 1.0.

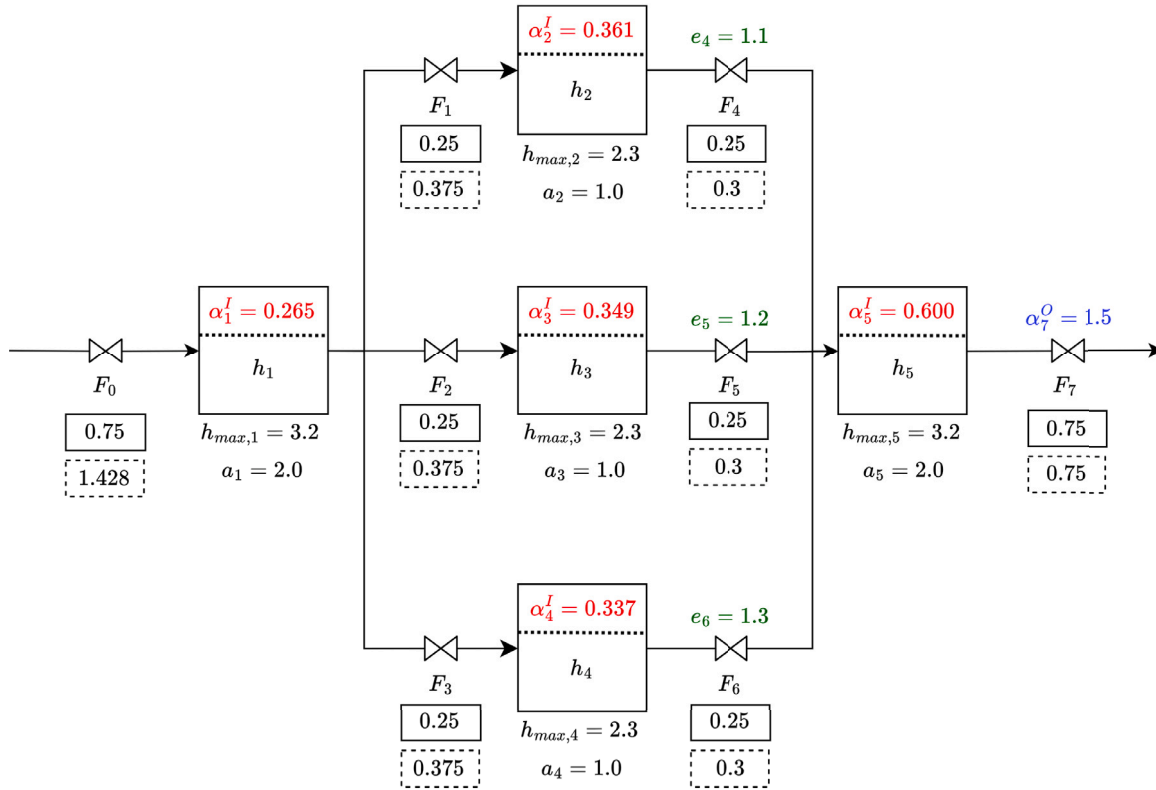


Fig. A.11. Example 5, tank arrangement with flow split and merger. Maximum flow rates in box with dashed lines, steady-state flow rates in box with solid lines.

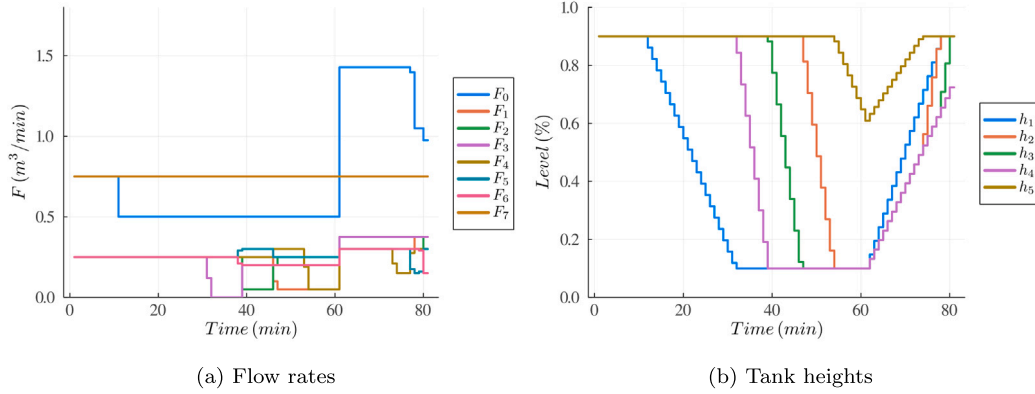


Fig. A.12. MPC Response, Example 5 - tank arrangement with flow split, flow merger and applied input regularisation of 10^{-3} .

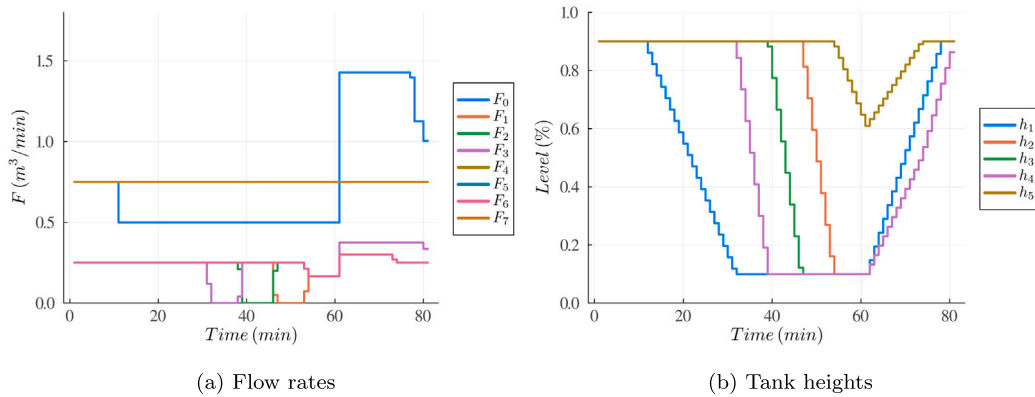


Fig. A.13. MPC Response, Example 5 - tank arrangement with flow split, flow merger and soft equality constraints for F_4 , F_5 , and F_6 .

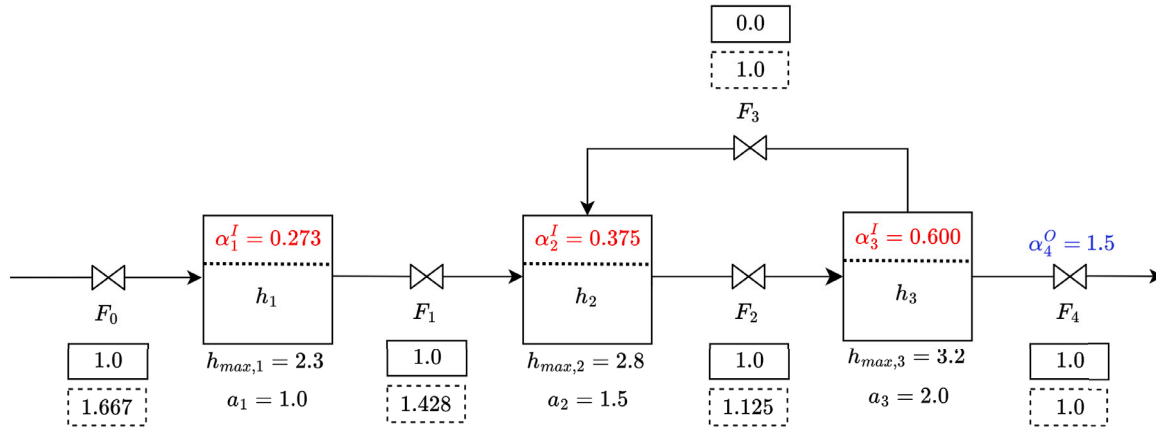


Fig. A.14. Example 6, tank arrangement with recycle stream. Maximum flow rates in box with dashed lines, steady-state flow rates in box with solid lines.

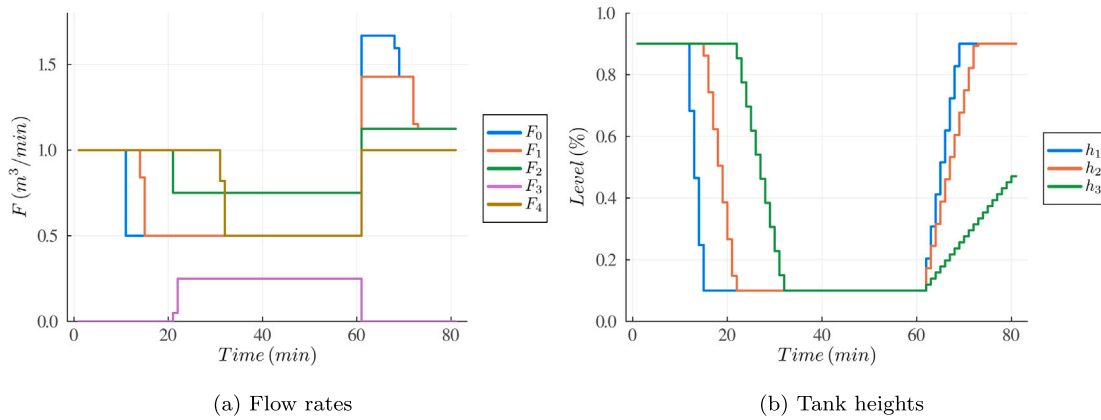


Fig. A.15. MPC Response, Example 6 - tank arrangement with recycle stream.

Table B.3

Example 3, parameters.

Tank Areas, m^2	[2.0 2.0 5.0 2.0 2.0]
Tank Heights, m^2	[5.0 5.0 6.0 4.0 4.0]
Maximum Flow Rates, F_{max} , m^3/min	[1.8 25.0 1.8 25.0 45.0 40.5 40.5 20.0 16.2 20.0]
Steady State Flow Rates, F_{ss} , m^3/min	[1.8 21.9 1.8 21.9 37.033 33.33 33.33 20.0 13.33 13.33]
Edge Weights, e	$e_2 = 1.2$, $e_9 = 1.8$
Outflow Weights, α^O	$\alpha_9^O = 1.5$
Inventory Weights, α^I	[0.170 0.176 0.214 0.375 0.155]

Table B.4

Example 4, parameters.

Tank Areas, m^2	1.0
Tank Heights, m^2	2.3
Max. Flow, F_{max} , m^3/min	[1.667 1.667 1.428 1.428 0.375 0.375 0.375 0.3 0.3 0.5 1.0 1.0]
Initial Flow, F_{ss} , m^3/min	[0.4 0.4 0.4 0.4 0.25 0.25 0.3 0.25 0.25 0.3 0.5 0.0 0.0]
Edge Weights, e	$e_2 = 1.2$, $e_7 = 1.1$, $e_8 = 1.2$, $e_{12} = 1.2$
Outflow Weights, α^O	$\alpha_9^O = 1.5$, $\alpha_{10}^O = 1.2$
Inventory Weights, α^I	[0.259 0.273 0.375 0.341 0.330 0.600 0.545 0.197]

Table B.5

Example 5, parameters.

Tank Areas, m^2	[2.0 1.0 1.0 1.0 2.0]
Tank Heights, m^2	[3.2 2.3 2.3 2.3 3.2]
Maximum Flow Rates, F_{max} , m^3/min	[1.428 0.375 0.375 0.375 0.300 0.300 0.300 0.750]
Steady State Flow Rates, F_{ss} , m^3/min	[0.75 0.25 0.25 0.25 0.25 0.25 0.25 0.75]
Edge Weights, e	$e_4 = 1.1$, $e_5 = 1.2$, $e_6 = 1.3$
Outflow Weights, α^O	$\alpha_7^O = 1.5$
Inventory Weights, α^I	[0.265 0.361 0.349 0.337 0.600]

Table B.6

Example 6, parameters.

Tank Areas, m^2	[1.0 1.5 2.0]
Tank Heights, m^2	[2.3 2.8 3.2]
Maximum Flow Rates, F_{max} , m^3/min	[1.667 1.428 1.125 1.000 1.000]
Steady State Flow Rates, F_{ss} , m^3/min	[1.0 1.0 1.0 0.0 1.0]
Edge Weights, e	–
Outflow Weights, α^o	$\alpha_4^o = 1.5$
Inventory Weights, α^I	[0.273 0.375 0.600]

Data availability

No data was used for the research described in the article.

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